**DO NOW — RECALL**

# Arrays in C#

Name: _____ Class: _____ Date: _____

**1. For a 3 × 3 array called board , what do GetLength(0) and GetLength(1) return?**

**2. What is the difference between a function and a procedure?**

**3. Convert hexadecimal 2F to denary.**

**4. Why must every value in a C# string[,] array be stored as a string?**

**DO NOW — CHECK / OBJECTIVES**

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

3 and 3. Dimension 0 is rows; dimension 1 is columns.

### 2. Model answer

A function returns a value; a procedure does not.

### 3. Model answer

47. 2 × 16 + 15 = 47 .

### 4. Model answer

An array has one declared element type, so every element must be compatible with that type.

### By the end, you can…

• declare and access one-, two- and three-dimensional arrays;

• use zero-based indexes and GetLength() correctly;

• traverse arrays using loops and choose suitable dimensions for stored data.

### Vocabulary — know these terms

Recall the term — Array

Recall the term — Element

Recall the term — Index

Recall the term — Dimension

Recall the syntax — []

Recall the syntax — [,]

Recall the syntax — [,,]

Recall the method — GetLength()

NEW LEARNING

# Read, annotate and explain

### Arrays up to 3 dimensions

string[] names;
string[,] results;
int[,,] readings;

Core rule: one array has one element type. More dimensions mean more indexes are needed to locate one element.

### 1D · one index

Indexes start at 0, so students[2] is Charlie .

### 2D · row and column

[0,0] means row 0, column 0. It outputs Amelia .

### GetLength() names the dimension

For the array above, the answers are 3 and 2 .

### 3D · layer, row, column

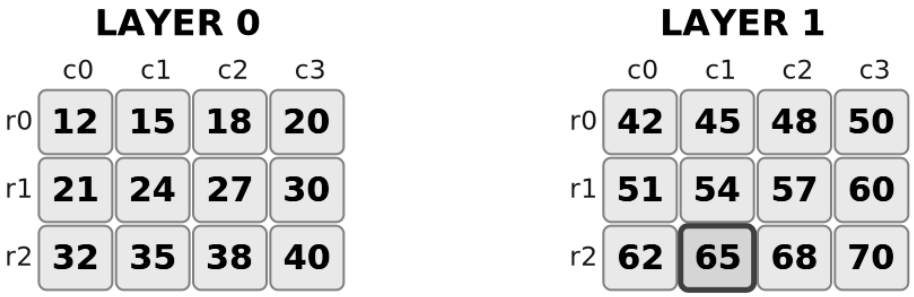
Read the indexes as layer → row → column . A 3D array is easiest to visualise as several 2D tables stacked together.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

### Arrays up to 3 dimensions




**scores[1,2,1] = 65**

A three-dimensional array as two layers of three rows and four columns; scores at layer one, row two, column one is 65

I DO — WORKED EXAMPLE

# Follow the worked method

### Build studentResults

### Rows are students · columns are fields

```
row 0 → Amelia | 72
row 1 → Ben    | 65
row 2 → Charlie| 81
```

We need 3 rows and 2 columns.

### Create the 2D array

```
string[,] studentResults =
{
    { "Amelia", "72" },
    { "Ben", "65" },
    { "Charlie", "81" }
};
```

Both columns use strings in this teaching example.

### Use row, then column

```
Console.WriteLine(studentResults[0,0]);
Console.WriteLine(studentResults[0,1]);
```

The outputs are Amelia and 72 .

### Visit every row

```
for (int row = 0;
     row < studentResults.GetLength(0);
     row++)
{
    Console.WriteLine(
        studentResults[row,0] + " " +
        studentResults[row,1]);
}
```

### Think in layers

```
LAYER 0
          col 0  col 1  col 2  col 3
row 0       12     15     18     20
row 1       21     24     27     30
row 2       32     35     38     40

LAYER 1
          col 0  col 1  col 2  col 3
row 0       42     45     48     50
row 1       51     54     57     60
row 2       62     65     68     70
```

Worked example: find scores[1,2,1] .

### Three dimensions need three loops

I DO — WORKED EXAMPLE

# Follow the worked method · continued

```
for (int layer = 0;
     layer < scores.GetLength(0);
     layer++)
{
    for (int row = 0;
         row < scores.GetLength(1);
         row++)
    {
        for (int column = 0;
             column < scores.GetLength(2);
             column++)
        {
            Console.WriteLine(
                scores[layer,row,column]);
        }
    }
}
```

**WE DO — GUIDED PRACTICE**

# Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

### Guided 1 — TRY: Read an indexed value

What is stored at index 2, and how many elements are in the array?

Method / working:

### Guided 2 — TRY: Read row and column

Find studentResults[2,1] and the two dimension lengths.

Method / working:

### Guided 3 — TRY: Find the address

The value 57 is in layer 1. What complete array reference accesses it?

Method / working:

### Guided 4 — TRY: Declare towns

OCR describes a 2D array called towns with 8 rows and 3 columns. Write a suitable declaration.

Method / working:

# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

### Guided 1: Read an indexed value CODE: string[] names = { "Amelia", "Ben", "Charlie", "Daisy" }; What is stored at index 2, and how many elements are in the array? Read the array

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 2: Read row and column CODE: string[,] studentResults = { { "Amelia", "72" }, { "Ben", "65" }, { "Charlie", "81" } }; Find studentResults[2,1] and the two dimension lengths.

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 3: Find the address CODE: LAYER 1 col 0 col 1 col 2 col 3 row 0 42 45 48 50 row 1 51 54 57 60 row 2 62 65 68 70 The value 57 is in layer 1. What complete array reference accesses it?

DISCUSS — Compare approaches, walk through one test input, and explain how you would detect or correct a mistake.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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### Guided 4: Declare towns OCR describes a 2D array called towns with 8 rows and 3 columns. Write a suitable declaration.

DISCUSS — Compare the representation or steps you used. Explain one decision and challenge one possible misconception.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

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YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

### Question 1

Find 24

Using the supplied scores array, change the Console.WriteLine statement so the program outputs 24 .

### Question 2

Find 38

Write a Console.WriteLine statement that outputs the value 38 .

### Question 3

Find the last value

Output the last value in the array without changing the array.

### Question 4

Change a value

Change the value 35 to 99 , then output it to prove that the value has changed.

### Question 5

Measure the array

Output the size of all three dimensions using scores.GetLength(0) , scores.GetLength(1) and scores.GetLength(2) .

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

### Question 6

Output one row
Use a for loop to output 12, 15, 18 without writing three separate Console.WriteLine statements.

### Question 7

Output the first layer
Use two nested loops to output all six values in the first layer: scores[0, ... , ...] .

### Question 8

Traverse the whole array
Use three nested loops and GetLength() to output every value in the 3D array.

### Question 9

Calculate the total
Adapt your three nested loops to calculate and display the total of every value in scores .

### Question 10

Find the largest value
Use three nested loops to find and display the largest value in the entire 3D array.
Do not assume 48 is the largest. Your algorithm must still work if the values in the array are changed.

**EXIT TICKET / REFLECTION**

# Check your understanding

Complete the actual lesson Exit Ticket before checking.

**int[,,] data = new int[2,3,4]; declares a three-dimensional array.**

**Circle: Confident / Getting there / Need help**

Explain what you can now do independently.

**GREEN PEN — Correct and explain**

Choose one error. Write the correct method and explain precisely why your first response was wrong.

**WE DO — CHECK AND IMPROVE**

# Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

### Guided 1 — CHECK

Read the array

### Guided 2 — CHECK

Locate the element

### Guided 3 — CHECK

Say it before you write it
Locate 57

### Guided 4 — CHECK

Build the declaration
OCR's mark scheme awards the declaration for identifying an array/string named towns and setting the two dimensions to 8 and 3.